

KIMBERLEY, South AfricaWMO: **684380**Lat: **28.8061S**Lon: **24.7697E**Elev: **3924**StdP: **12.73**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.1	30.9	4.6	8.1	58.4	8.7	9.9	57.2	24.8	59.9	22.5	62.3	2.6	10	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.7	97.1	61.8	95.0	61.6	93.0	61.5	70.0	82.7	68.7	81.3	67.5	80.0	11.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.7	113.8	73.3	65.4	108.7	72.5	64.1	103.8	72.0	36.7	81.8	35.4	81.3	34.4	80.1	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	20.4	18.1	22.1	100.3	2.5	2.0	20.3	101.8	18.8	102.9	17.4	104.0	15.6	105.5		
(5)			19.9	73.1	2.2	5.0	18.3	76.8	16.9	79.7	15.7	82.5	14.1	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	76.6	75.9	72.1	64.4	57.5	51.4	50.9	55.9	62.8	68.7	72.2	75.2		
	DBStd	11.02	4.82	4.85	5.51	6.22	6.21	5.79	6.43	7.36	7.67	6.82	6.66	5.02		
	HDD50	170	0	0	0	1	15	52	69	29	4	0	0	0		
	HDD65	1665	1	2	10	84	237	409	437	292	130	43	19	1		
	CDD50	5727	826	724	685	432	248	93	97	212	387	578	665	780		
	CDD65	1746	361	306	230	64	5	0	0	10	63	157	233	317		
	CDH74	21050	4069	3208	2385	761	184	13	32	267	1108	2180	3043	3800		
	CDH80	9627	2064	1543	1018	188	12	0	2	36	402	951	1484	1927		
(14) Wind	WSAvg	8.0	9.2	8.2	7.6	6.4	6.2	6.4	7.0	8.0	8.4	9.2	9.7	9.7		
(15) Precipitation	PrecAvg	16.0	2.6	2.8	2.5	1.7	0.6	0.3	0.2	0.3	0.4	1.3	1.7	1.9		
	PrecMax	30.0	10.2	13.1	6.1	6.3	3.2	2.7	1.6	2.7	2.8	6.3	5.6	6.3		
	PrecMin	7.3	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0		
	PrecStd	5.6	2.0	2.1	1.5	1.4	0.8	0.5	0.4	0.5	0.7	1.3	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.9	97.7	95.8	89.7	82.0	75.6	77.6	83.8	92.1	96.8	98.1	100.5		
		MCWB	62.4	64.3	62.5	61.1	56.0	53.4	51.9	54.4	58.2	60.4	60.2	61.6		
	2%	DB	96.7	95.1	92.1	84.4	78.7	72.0	73.3	80.2	88.3	92.3	94.9	96.6		
		MCWB	61.9	62.9	62.7	59.5	54.5	51.6	50.2	52.9	56.8	58.6	60.5	61.0		
	5%	DB	93.6	92.5	89.2	81.0	75.9	69.7	70.2	76.9	84.6	89.4	92.3	93.3		
		MCWB	62.2	63.7	62.4	59.2	54.1	50.3	49.3	51.9	54.8	58.3	59.9	61.0		
	10%	DB	91.0	89.5	86.0	78.4	72.9	66.8	67.1	73.2	80.8	85.8	89.0	90.2		
		MCWB	62.9	64.0	62.3	58.4	54.0	49.1	48.4	51.1	53.9	57.4	59.2	61.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.4	74.6	70.1	66.2	60.9	55.8	54.8	58.5	62.0	67.2	69.4	69.5		
		MCDB	84.6	82.6	81.1	77.8	72.4	69.9	71.9	75.3	82.5	78.2	80.0	83.6		
	2%	WB	69.5	70.7	68.3	64.1	58.2	53.8	52.3	55.8	59.5	64.0	66.4	67.7		
		MCDB	82.3	83.0	79.5	76.5	70.9	67.6	69.1	73.2	78.9	80.8	80.1	80.8		
	5%	WB	68.3	68.9	66.8	62.7	56.5	52.0	50.6	53.9	57.7	62.0	64.6	66.3		
		MCDB	80.8	82.1	79.1	74.2	70.1	66.1	67.1	71.4	77.7	79.9	79.4	80.0		
	10%	WB	67.0	67.5	65.3	61.2	55.0	50.4	49.0	51.9	55.9	60.2	63.0	65.0		
		MCDB	79.7	80.2	78.1	72.2	68.8	64.0	65.2	69.9	75.7	77.9	78.9	79.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.7	24.9	25.1	25.7	28.0	29.1	30.0	30.9	31.3	29.9	29.0	27.7		
		MCDBR	30.5	29.2	29.5	28.2	31.6	31.9	32.4	33.7	35.0	33.7	32.7	32.4		
		MCWBR	10.5	10.3	10.4	11.0	14.2	16.3	16.5	16.5	16.1	13.9	12.9	12.5		
	5% WB	MCDBR	23.0	23.0	23.1	22.8	26.3	27.9	29.5	29.5	30.0	27.7	25.7	25.6		
		MCWBR	8.8	9.8	9.0	10.5	13.1	15.1	16.0	15.5	14.7	12.3	11.4	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.349	0.340	0.307	0.284	0.267	0.265	0.310	0.352	0.363	0.322	0.326		
	taud		2.475	2.450	2.452	2.520	2.562	2.579	2.566	2.418	2.291	2.294	2.469	2.497		
	Ebn at Noon		318	309	302	297	291	290	295	291	293	301	321	323		
	Edh at Noon		37	37	35	30	25	23	25	32	41	43	37	37		
(44) All-Sky Solar Radiation	RadAvg		2383	2150	1900	1545	1295	1158	1261	1529	1895	2209	2470	2534		
	RadStd		161	180	100	79	59	50	40	75	82	102	122	118		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	+1.22	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page